

Three-Window Dynamics in Phase-Locked Geometric Structures: A Band-Pass and Orientation-Torque Framework

Abstract

In many discrete and anisotropic systems, observable responses do not vary continuously with control parameters such as driving speed, distance, or orientation. Instead, responses often appear only within limited operational ranges and at specific configurations. In this work, we propose a phenomenological framework—referred to as Three-Window Dynamics—to describe such behaviors in phase-locked geometric structures. The framework identifies three necessary constraints for observable responses: a band-pass frequency window, a finite interaction-distance window, and a discrete orientation window. We further show that the dominant macroscopic behavior is governed by orientation torque and discrete configuration switching rather than continuous attractive or repulsive forces. This framework does not introduce new fundamental interactions and provides testable predictions for future experimental studies.

1. Introduction

Geometric structures with discrete symmetry appear across physics, engineering, and materials science. In many such systems, responses to external modulation exhibit anisotropic and non-continuous behavior. Conventional continuous models often fail to capture these effects.

2. Three-Window Constraint

2.1 Frequency Window

Responses exhibit a band-pass profile characterized by an optimal driving rate ω^* . Low speeds yield weak excitation, intermediate speeds maximize response, and high speeds cause phase mismatch.

2.2 Distance Window

Observable responses occur only within a finite interaction shell, with saturation at short distances and decay at long distances.

2.3 Orientation Window

Discrete symmetry leads to preferred orientations where responses are enhanced.

3. Discrete Phase Switching

Responses emerge as transient enhancements at specific orientations, interpreted as configuration switching.

4. Orientation-Torque Dominated Behavior

The dominant macroscopic effect is orientation torque rather than radial attraction or repulsion.

5. Unified Phenomenological Model

The framework unifies frequency, distance, and orientation constraints through an effective potential landscape.

6. Testable Predictions

- (1) Band-pass frequency response
- (2) Finite distance interaction shell
- (3) Discrete stable orientations

7. Conclusion

Three-Window Dynamics provides a unified framework for understanding non-continuous responses in phase-locked geometric structures.